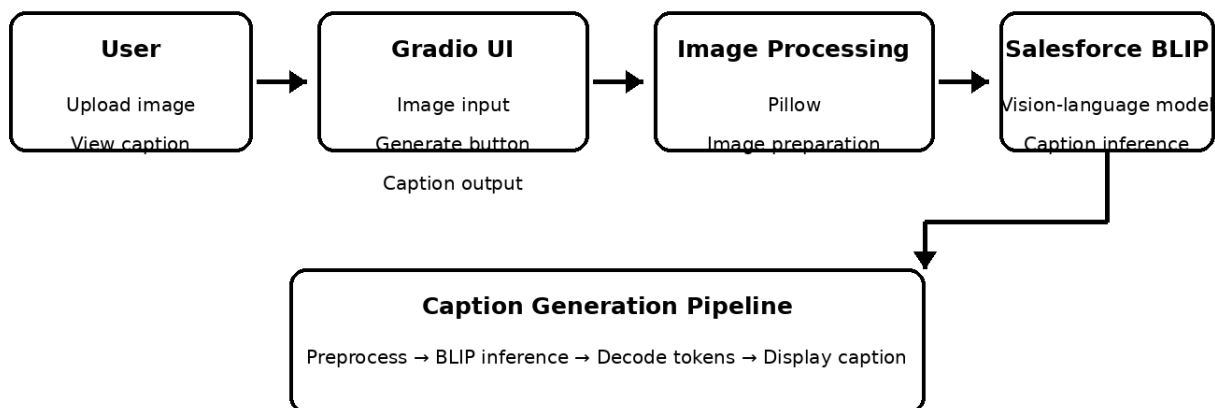


AI-BASED IMAGE CAPTIONING SYSTEM

CodSoft Artificial Intelligence Internship — Task 3

Project Report

AI Image Captioning System — System Architecture



1. Introduction

Image captioning is a computer vision and natural-language processing task in which an artificial intelligence system analyzes an image and produces a textual description. This project implements a practical image captioning application using a pretrained vision-language model.

2. Objective

The objective is to build an easy-to-use application that converts visual information from an uploaded image into a meaningful natural-language caption while demonstrating image processing, deep learning inference, and an interactive web interface.

3. Problem Statement

Images contain rich visual information, but computers require machine-learning techniques to interpret that information in human-readable form. The project addresses the problem of automatically generating a concise textual description from an input image.

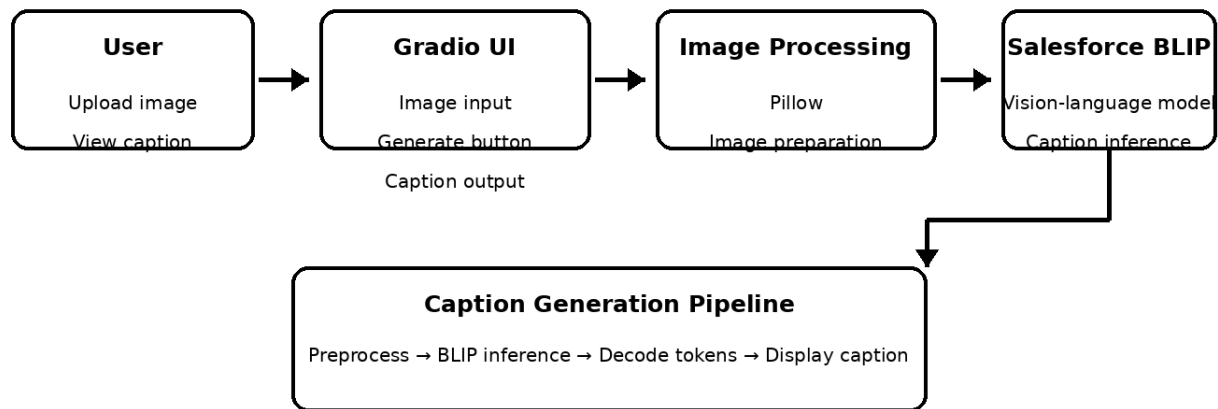
4. Proposed Solution

The system uses Pillow for image handling, a pretrained Salesforce BLIP image-captioning model through Hugging Face Transformers, PyTorch for inference, and Gradio for the interactive interface. The model is used for inference rather than custom training.

5. System Architecture

User input → Gradio interface → image processing → Salesforce BLIP → caption generation → result display.

AI Image Captioning System — System Architecture



6. Technologies Used

Technology	Purpose
Python 3.12	Application development and execution
PyTorch	Deep learning inference framework
Hugging Face Transformers	Pretrained model integration
Salesforce BLIP	Image caption generation
Pillow	Image loading and preprocessing
Gradio	Interactive web user interface

7. System Requirements

Software: Python 3.12, virtual environment, PyTorch, Hugging Face Transformers, Pillow, Gradio, and a supported operating system.

Hardware: Modern CPU, adequate RAM and disk space; GPU optional.

8. Methodology / Workflow

1. User uploads an image through the Gradio interface.
2. The image is loaded and prepared using Pillow.
3. The processed image is passed to the pretrained Salesforce BLIP model.
4. BLIP performs vision-language inference and generates caption tokens.
5. The generated tokens are decoded into natural-language text.
6. The caption is displayed to the user.

9. AI Model — Salesforce BLIP

The application uses Salesforce BLIP through Hugging Face Transformers. BLIP is a pretrained vision-language model used here to connect visual information with natural-language caption generation. The model is loaded locally and performs inference on uploaded images.

10. Image Processing

Pillow is used to load and prepare images before model inference. The application interface supports JPG, JPEG, PNG and WEBP images.

11. Application Implementation

app.py — Main Gradio application and UI

caption_generator.py — Model loading and caption generation

image_processor.py — Image preparation and preprocessing

utils.py — Reusable helper functionality

test_caption.py — Model/application testing

requirements.txt — Python dependencies

assets/ — Test and sample images

screenshots/ — Application screenshots

models/ — Reserved model resource directory

app.py launches the Gradio application and connects the image input to the caption-generation pipeline. The interface includes image upload, Generate Caption, caption output, and Clear functionality.

12. User Interface

The Gradio interface provides a straightforward workflow: upload an image, generate a caption, and view the AI-generated description. The UI also presents model and technology information.

13. Testing

Test	Expected Result	Observed Result
JPG/JPEG upload	Image accepted	Successful
PNG/WEBP upload	Image accepted	Supported
Generate caption	Caption displayed	Successful
Bridge/temple image	Scene described	Successful
Portrait image	Subject/background described	Successful
Model initialization	Model loads	Successful on CPU

14. Sample Results

For the bridge-and-temple test image, the model generated: **“a woman standing on a bridge in front of a temple”**

For a portrait-style test image, the system generated a caption describing a woman standing in front of a white backdrop.

15. Results

The completed system successfully loads the pretrained model, accepts image input, performs image-captioning inference on the CPU, and displays generated captions through the Gradio interface.

16. Advantages

- Simple upload-and-generate workflow.
- Uses a pretrained vision-language model instead of training from scratch.
- Runs locally through a Gradio interface.

- Supports JPG, JPEG, PNG and WEBP images.
- Produces human-readable descriptions of major visual content.

17. Limitations

- Caption quality depends on the pretrained model and image quality.
- CPU inference can be slower than GPU execution.
- The model may miss fine-grained or ambiguous scene details.
- The initial pretrained-model download requires substantial disk space and setup time.

18. Future Enhancements

- Add GPU acceleration and inference optimization.
- Allow beam-search or generation-parameter configuration.
- Support multilingual caption generation.
- Add caption history and export options.
- Deploy the application as a hosted web service.
- Evaluate caption quality on benchmark datasets using standard metrics.

19. Conclusion

The AI-Based Image Captioning System demonstrates an end-to-end application of artificial intelligence for converting visual content into natural-language descriptions. By combining Pillow, PyTorch, Hugging Face Transformers, Salesforce BLIP, and Gradio, the project demonstrates practical concepts in computer vision, vision-language models, preprocessing, inference, and AI application development.

20. References

- Hugging Face Transformers documentation and model ecosystem.
- Salesforce BLIP image-captioning pretrained model: [Salesforce/blip-image-captioning-base](#).
- PyTorch documentation.
- Pillow documentation.
- Gradio documentation.

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